

**Unit:** 6 Unit 4: Absolute Value

**Name:** \_\_\_\_\_

**Topic:** Absolute Value Function and its Graph

**Date:** \_\_\_\_\_

### Challenge (Habanero)

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#### Open Ended Questions (FRQ)

**Q9.** Graph the function  $y = |x + 1| - 3$  and identify the vertex, axis of symmetry, and direction of opening. Show all work.

**Q10.** A construction company is building a new skyscraper with a unique design. The height  $y$  of the building above the ground is given by  $y = |x - 10| + 5$ , where  $x$  is the distance from the center of the building. Find the  $x$ -coordinate of the vertex of the building and the maximum height of the building. Show all work and explain your answer.

#### Chef's Tips

- Chef's Tips: Remember to use the definition of absolute value to solve equations and inequalities.
- Chef's Tips: When graphing absolute value functions, identify the vertex and axis of symmetry.
- Chef's Tips: Pay attention to the direction of opening of the graph.